

# File and Terminal

the file system, paths, and talking to the computer directly

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## File System & Directory Structure

Files are not scattered randomly across a drive. They sit in a tree, and every file has an exact address inside that tree.

```
C:\
├── Users
│   └── Student
│       ├── Documents
│       │   └── notes.txt
│       └── Downloads
└── Program Files
```

Reading that structure outward, the hierarchy is: **File** → **Folder** → **Directory** → **Drive**.

## Paths

A path is the address of a file — how you tell the computer exactly where something lives.

ABSOLUTE PATH

```
C:\Users\Student\Documents\notes.txt
```

### Absolute Path

The complete location, starting from the drive. Works from anywhere.

### Relative Path

The location relative to the directory you are currently in.

File extensions tell you the type: `.txt` `.jpg` `.cpp` `.exe`. Hidden files and separate drives (C:, D:) all follow the same rules.

## Terminal — GUI vs CLI

There are two ways to tell a computer what to do.

### GUI

Graphical User Interface. You interact using visual elements — windows, icons, buttons and menus.

Click → Icon → Folder → File

### CLI

Command Line Interface. You interact by typing commands.

Type Command → Terminal → File / Program

Neither one is better. A GUI is easier to learn. A CLI is faster and repeatable once you know it — which is why developers live in the terminal.

## Everyday Commands

| ACTION            | MACOS /<br>LINUX | WINDOWS POWERSHELL        |
|-------------------|------------------|---------------------------|
| Current directory | <code>pwd</code> | <code>Get-Location</code> |
|                   |                  |                           |

| ACTION              | MACOS /<br>LINUX        | WINDOWS POWERSHELL                                 |
|---------------------|-------------------------|----------------------------------------------------|
| List files          | <code>ls</code>         | <code>Get-ChildItem</code>                         |
| Change<br>directory | <code>cd folder</code>  | <code>Set-Location folder</code>                   |
| Create folder       | <code>mkdir test</code> | <code>New-Item test -ItemType<br/>Directory</code> |



The commands differ between systems, but the four **actions** are identical everywhere: know where you are, see what is there, move somewhere else, make something new.

## What Happens When You Type a Command?

Say you type `python` in the terminal and press enter. How does the computer know where the Python program lives on your disk?

Terminal — you type the command



OS searches the PATH



Executable is found



Program runs

#### DEFINITION

**PATH** is a list of directories where the operating system looks for executable programs when you type a command.



This is why "command not found" happens even when the program **is** installed — the OS simply does not know which folder to look in. Adding that folder to PATH fixes it. You will hit this exact problem when you set up your C++ compiler.

#### SUMMARY

- File → Folder → Directory → Drive.
- Absolute path starts from the drive; relative path starts from where you are.
- GUI uses visuals; CLI uses typed commands.
- PATH is the list of folders the OS searches when you type a command.